

1. Cellular microtubules can behave as qubits and even store information with memory.
2. These qubits can occupy a superposition of states and certain associated proteins in microtubules can, via synaptic impulses, 'tune' these states to 'orchestrate' coherence in an organized manner.
3. These 'orchestrated' coherent states can influence neural impulses from within the neuron, leading to a 'conscious' decision or state of mind.

The theory likens the brain to a functioning quantum computer. Another analogy might help here. Think of the qubits as independent, talented musicians in an orchestra. They can play any instrument and any piece of music. If they do so randomly, it results in cacophony. Even if they play the right instrument and piece of music, unless each group plays its part at the right time, there is still no orchestra. Enter the conductor who 'orchestrates' the collection of musicians to produce music.

As wacky as it may sound some research groups have tried to corroborate the theory by carrying out experiments. As recently as 2013, a group found out EEG rhythms derive from microtubule vibrations indicating that may be, at some level, quantum weirdness in microtubules has some role in modulating brain waves. Another group claimed that anesthesia that is responsible for controlling consciousness in particular part acts via microtubules. These results, while encouraging do not, however, mention any events quantum in nature. Then there is the icky problem of quantum – decoherence.

The effect of quantum coherence more often than not occurs at low temperatures. The brain with all its electronic impulses is warm – warm enough to be hostile against such quantum events. Even though evidence of quantum effects in warm environments, most notably in photosynthesis in plants, has been pointed out, there has been some damning criticism against Orch-OR – none more than a paper published in 2000. The author of the study, physicist Max Tegmark calculated that interactions

between atoms in a coherent state and a neuron would produce enough noise to result in a decoherence event. Calculations suggest that quantum coherence 'leaks' away much quicker than the time it must hold for driving and discharging signals. But what if aside from microtubules, another object could sustain coherence for a longer time?



British Physicist Roger Penrose famous for his work on black holes and now his controversial theory of consciousness

MORE QUANTUM GOODNESS

Matthew Fisher, a well-regarded scientist, (and son of Michael Fisher, a leading figure in statistical physics) also has a theory of consciousness. Matthew Fisher claims that phosphorus, abundant in living cells in the form of ions (with electrons lost or gained), in special 'entangled' states, can replace the qubits of Penrose's theory. Quantum objects in an entangled state are rigidly codependent, in the sense that the properties of one object strongly depends on the other. If these phosphate ions are in conjunction with calcium ions in a larger 'super molecule,' then it turns out entanglement saves them from decoherence. These molecules called Posner molecules, once entangled can resist decoherence for even a day, then becoming a candidate for affecting brain function at realistic timescales. Because this entanglement occurs via the spins of the individual nuclei, it is more resist-

ant to leakage via interactions with surroundings. Here's how this would work. Once the hypothetical Posner molecule enters a neuron, it would shed calcium and trigger an impulse. Because Posner molecules are entangled via spins, a collection of such signals can manifest like a coherent thought. Fisher's theory provides an elegant solution to the coherence decay problem posed by Penrose-Hameroff theory by replacing microtubules with Posner molecules as the fundamental quantum unit, nuclear spins being the seat for the brain's quantum processing. By combining concepts of organic biochemistry, neuroscience, and quantum mechanics, Fisher shows that phosphorus in a tailored state can form a stable biological qubit – in theory. The challenge now is to verify these claims experimentally, most notably check if Posner molecules can and are being formed in our nervous system.

IIT (NO, NOT THE COLLEGE)

Moving on from quantum mechanics, scientists have tried to ascertain if consciousness can be described by the one thing that defines the current century more than anything else – information. A modern theory born out of this endeavor is Integrated Information Theory (IIT). The two core postulates of this theory are:

1. Consciousness rather than being one unifying perception is made of a multitude of different frames filled with massive amount of information.
2. All this information is integrated and connected such that you cannot selectively ignore any of it.

At the heart of this theory lies the belief that the brain is deeply integrated, enough to perceive each frame of consciousness, all of its information and relate 'meaning' to it. For example, take the movie *Boyhood*, a movie set over a 12-year span in the life on a single boy. To a computer, the movie's constituent frames are all independent and disconnected. The computer cannot recognize that the bug toothed boy in the first half of the film is the lanky teenager in the second half, purely from looking at the frames. But